

Introduction to Pavement Engineering

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Course Notes

§ Course notes are available for free this year for enrolled students in the form of pdf files posted on Learn

§ Recommended Text Book:

§ Pavement Analysis and Design, Yang H. Huang

§ Pavement Design and Materials, Papagiannakis and Masad

§ Technical Basis of Austroads Guide to Pavement Technology Part 2: Pavement Structural Design

§ www.austroads.com.au

Classification: In-Confidence

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Lectures / Tutorials

§ Wednesday 2:00-3:00 pm, Link309

§ Thursday 3:00-4:50 pm, Link309

§ Friday 11:00-12:00 pm, Link309

§ (Lecture Theatre varies – see timetable)

§ **Tutorial:**

§ Friday 12:00-1:00PM Link309

§ Combined {20/3, 27/3, 24/4, 1/5, 8/5, 15/5, 22/5, 29/05}

Classification: In-Confidence

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Laboratory Schedule

Lab A				
Activity	Day	Time	Location	Date
01	Monday	12:00 - 13:30	Trans Lab E111	16 March
02		13:30 - 15:00	Trans Lab E111	16 March
Lab B				
01	Monday	12:00 - 13:30	Trans Lab E111	23 March
02		13:30 - 15:00	Trans Lab E111	23 March
Lab C				
01	Monday	12:00 - 13:30	Trans Lab E111	20 April
02		13:30 - 15:00	Trans Lab E111	20 April

4/5 lab is back plan for any emergency

Classification: In-Confidence

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Coursework and Due Dates

Assessment	Weight	Date
Quiz 1	%5.5	14/03
Quiz 2	%5.5	25/04
Quiz 3	%5.5	15/5
Quiz 4	%5.5	29/5
Mid Term Test (30%)	30%	030/4 @ 6:00 pm
Lab Report	8.0%	8/5
Final Exam	40%	TBD
ΣΣ	100%	

Classification: In-Confidence

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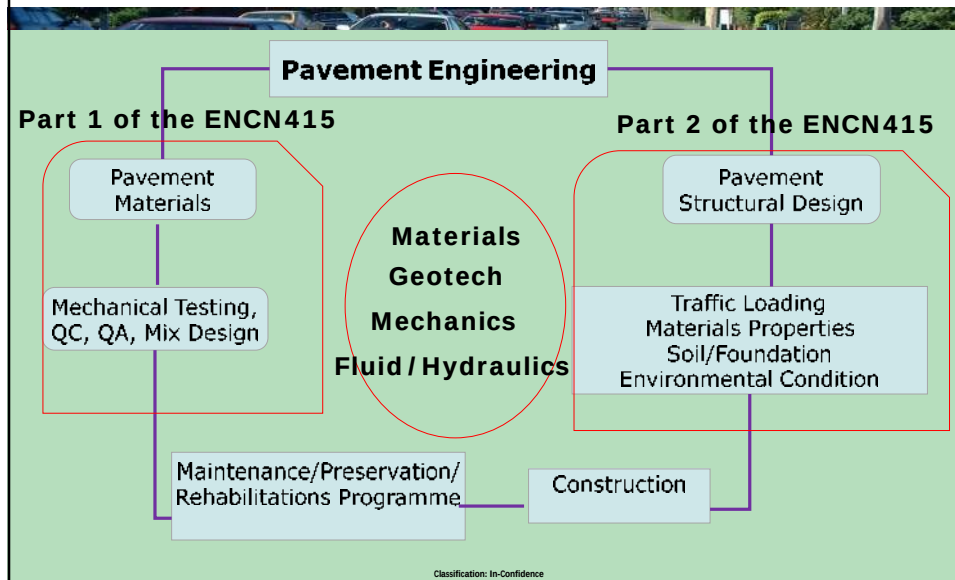
Health and Safety

- § H&S are very strictly followed
- § In the lab: You must wear your Safety Shoes or you will not be permitted in the lab
- § Other PPG will be available in the lab such as safety goggles, visible vests, long sleeve coats,

Classification: In-Confidence

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Pavement Engineering



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New Roads of National Significance

- § Northland Expressway: A preferred route has been confirmed, with benefits including up to 38 minutes faster travel between Te Hana and Whangārei, and a reduction in deaths and serious injuries.
- § East West Link (Auckland): This project aims to cut congestion time almost 30%, saving up to 11.6 minutes to SH1 and 3.2 minutes to SH20.
- § Hamilton Southern Links: Expected to reduce peak trip times by 14 minutes and support the development of 17,300 homes and 7,100 jobs by 2055.

Classification: In-Confidence

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- § Petone to Grenada and Cross Valley Link:
These projects are part of the ongoing efforts to improve connectivity and reduce congestion in key areas.
- § SH1 Wellington improvements: Including a new Mt Victoria tunnel, which is expected to enhance traffic flow and safety in the region

Classification: In-Confidence

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**Any pavement repair without
FIRST fixing the cause of the
problem (in this case drainage)
will not make any sustainable
solution**

10

Transmission Gully: Major Repairs



Classification: In-Confidence

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Pothole problem 'systemic' warns expert, as \$4b spread of funding revealed

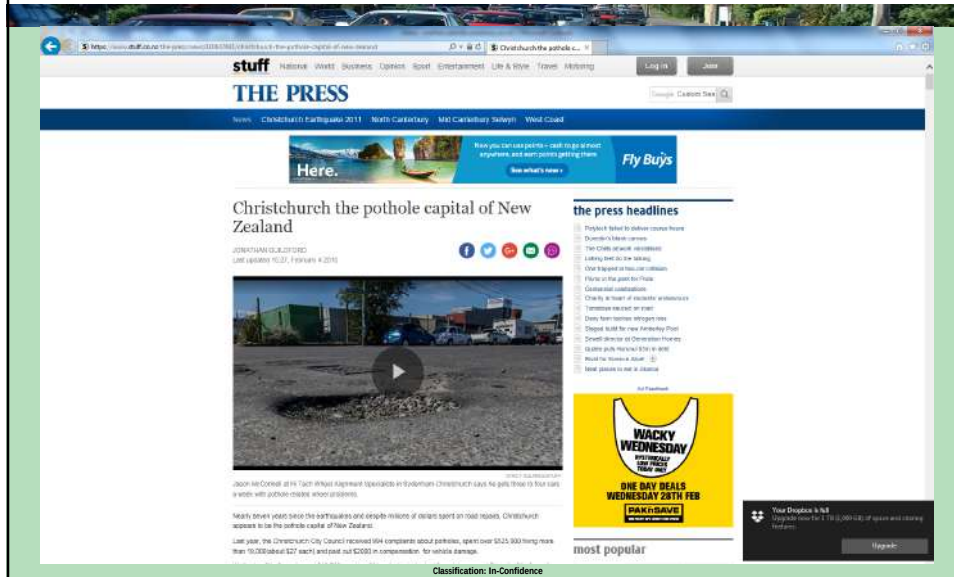


<https://www.thepress.co.nz/nz-news/350304005/pothole-problem-systemic-warns-expert-4b-spread-funding-revealed>

12

From the Press

<https://www.stuff.co.nz/the-press/news/100847641/christchurch-the-pothole-capital-of-new-zealand>



13

From the Press

§ In 2019, the Christchurch City Council received 994 complaints about potholes, spent over \$525,000 fixing more than 19,000 (about \$27 each) and paid out \$2000 in compensation for vehicle damage.

Classification: In-Confidence

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Public comments



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My comments to the Press



- § University of Canterbury associate engineering professor Mofreh Saleh said he had noticed many examples where the materials were applied and compacted incorrectly.
- § He said emergency repairs – where a pothole was simply filled with asphalt – was allowable as a temporary solution in winter months, but was against national code of practice when done in summer months.
- § "If it is done correctly it shouldn't come back," he said.

Classification: In-Confidence

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My comments to the Press

- § Saleh said in most other first world countries strict quality was enforced at each step of roading construction. New Zealand laws needed to catch up.
- § By spending about five per cent extra on a project to have quality control, less money would be spent on repairs, Saleh said.
- § "The science and knowledge are there and the expertise exists," he said

Classification: In-Confidence

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Is it only Domestic Issue?

Fixing potholes The hole story

Researchers are inventing new ways to prevent a motoring curse

POTHLES are a scourge of rich and poor countries alike. The American Automobile Association recently calculated that 16m drivers in the United States suffered pothole damage to their vehicles in the past five years. That damage ranged from punctures, via bent wheels, to broken suspensions. The bill to fix it was also at \$3 billion a year. In India, meanwhile, the cost of potholes is often paid in a harsher currency than dollars. There, more than 3,000

people a year are killed in accidents involving them. Yet cash-strapped governments often ignore the problem, letting roads deteriorate. In Britain, for example, some £2 billion (\$2.7 billion) would be needed to make all roads pothole-free. Ways of repairing potholes more cheaply and enduringly would thus be welcome. And several groups of researchers are working on it.

The most common cause of potholes is water penetrating cracks in a road's surface



From The Economist June, 11th, 2016

USA: Cost to fix damage to cars \$USD3 billion / year

India: 3000 people killed / year in accidents related to Potholes

UK: \$17 billion required to make pothole free roads

Classification: In-Confidence

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Seven Sharp (Potholes)

- § [Seven Sharp - Who should pay when you drive over a pothole on a state highway and blow out two tyres? | Facebook](#)

Classification: In-Confidence

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Waikato Expressway: NZTA won't say what's causing new motorways to crack

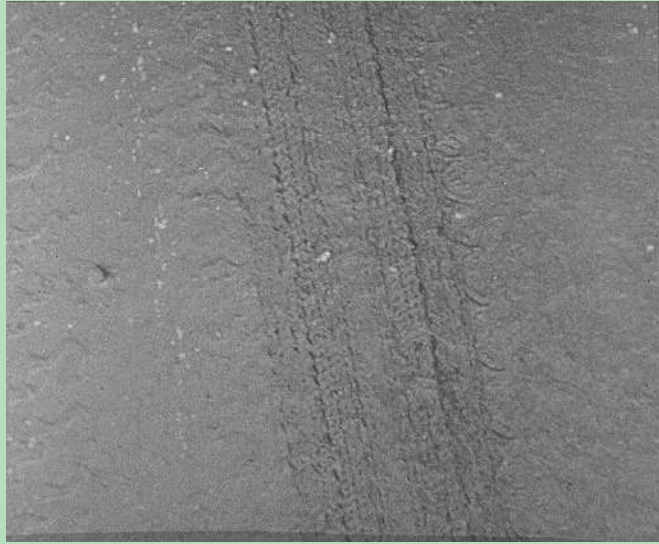
- § The New Zealand Transport Agency is refusing to say what it has found out about the cause of hugely **costly cracks and rutting** on expressways that are just a few years old.
- § The cost of damage to the surface of the Kōpiti Expressway and the Te Rapa and Ngaruawahia sections of the Waikato Expressway **has topped \$80 million**.
- § The massive bills do not stretch to long-term solutions in Waikato, covering only repairs and monitoring.

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Classification: In-Confidence

20

Bleeding / Flushing



21

Attempts to fix Transmission Gully made it worse



Extensive stretches of flushing on the southbound lanes, and some flushing on the northbound lanes.

Classification: In-Confidence

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Types of pavements

1. Flexible pavements

- a. Unbound gravel base/subgrade (chip sealed or thin asphalt layer)
- b. Structural Asphalt (> 100 mm of base/subbase)
- c. Structural asphalt over CTB base/subbase

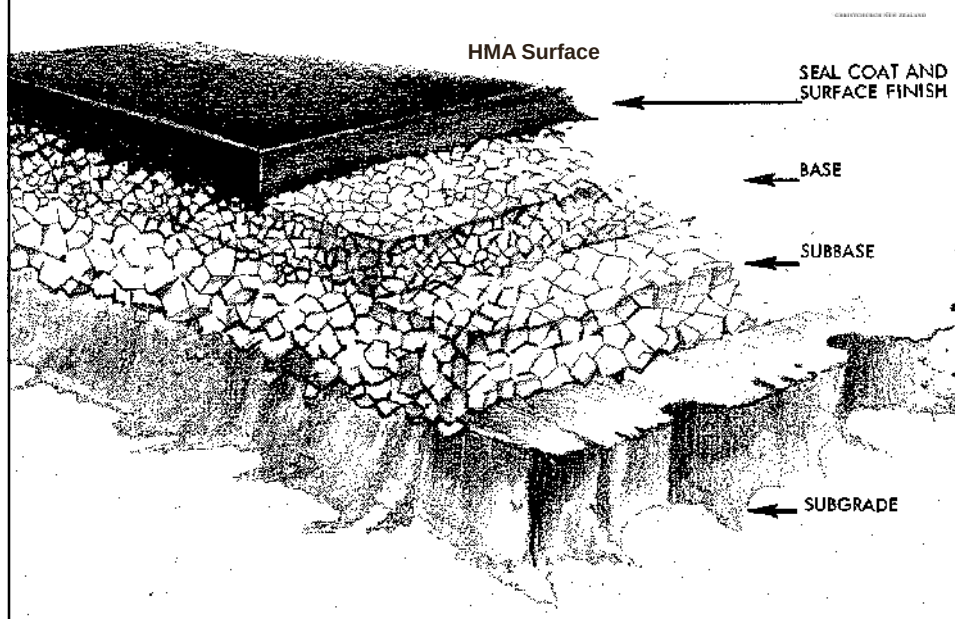
2. Rigid pavements

3. Composite pavements

Classification: In-Confidence

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Flexible Pavement



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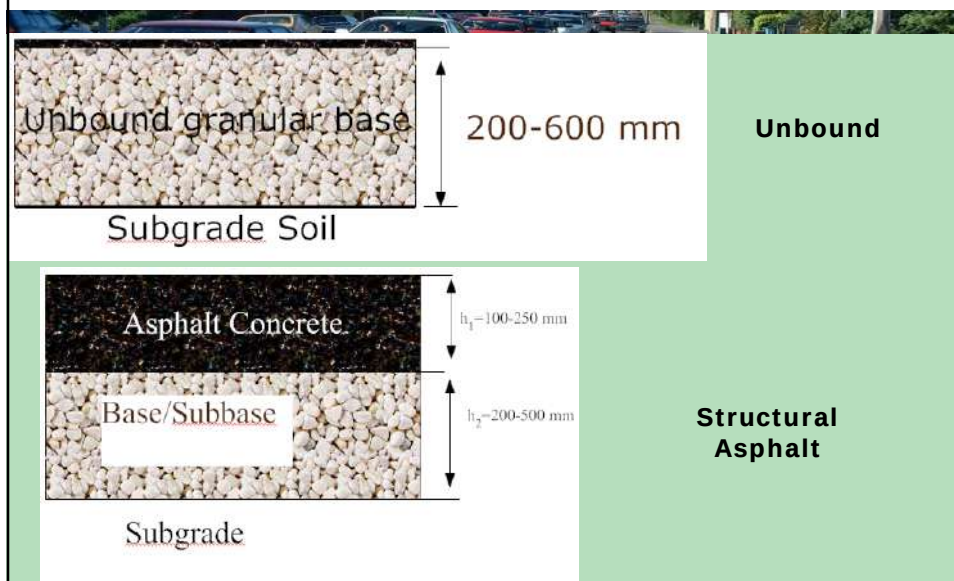
Pros / Cons

- § Asphalt roads are less durable
- § Asphalt roads are less expensive
- § More asphalt roads than concrete roads
- § Selection should be based on life-cycle cost

Classification: In-Confidence

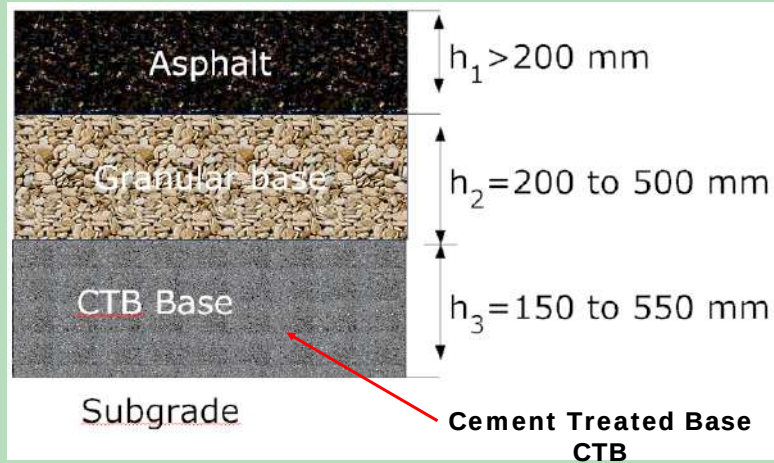
25

Different types of Flex Pavements



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CTB pavements (Upside Down pavements)



Classification: In-Confidence

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Flexible vs. Rigid Pavements

§ Load distribution

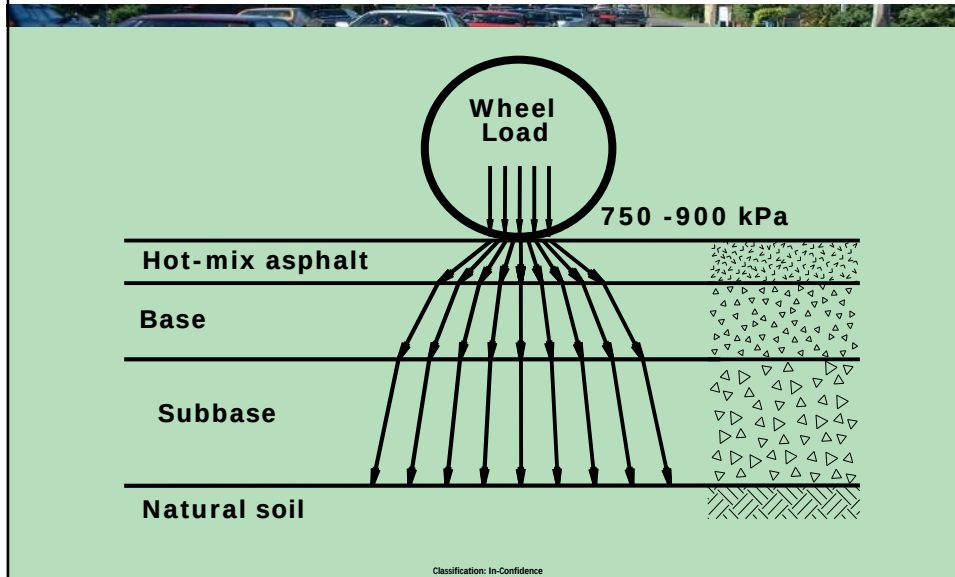
§ Initial cost

§ Durability

Classification: In-Confidence

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Distribution of Wheel Load



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Initial Cost

- § Flexible pavement has lower initial cost
 - But may not have lower life-cycle cost

Classification: In-Confidence

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Durability

§ Flexible pavement is less durable than rigid pavement

Classification: In-Confidence

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Function of Base / Subbase

§ Flexible Pavement

- Structural support
- Drainage
- Control of frost effect
- Reduce effect of volume change of subgrade

Classification: In-Confidence

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Factors Affecting Pavement Design / Performance



- Traffic
- Subgrade soil support
- Materials of construction
- Structural characteristics
- Construction and maintenance variation
- Environmental cond. Moisture/Temp.
- Maintenance / rehabilitation programs

Classification: In-Confidence

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Basic Distress Mechanisms

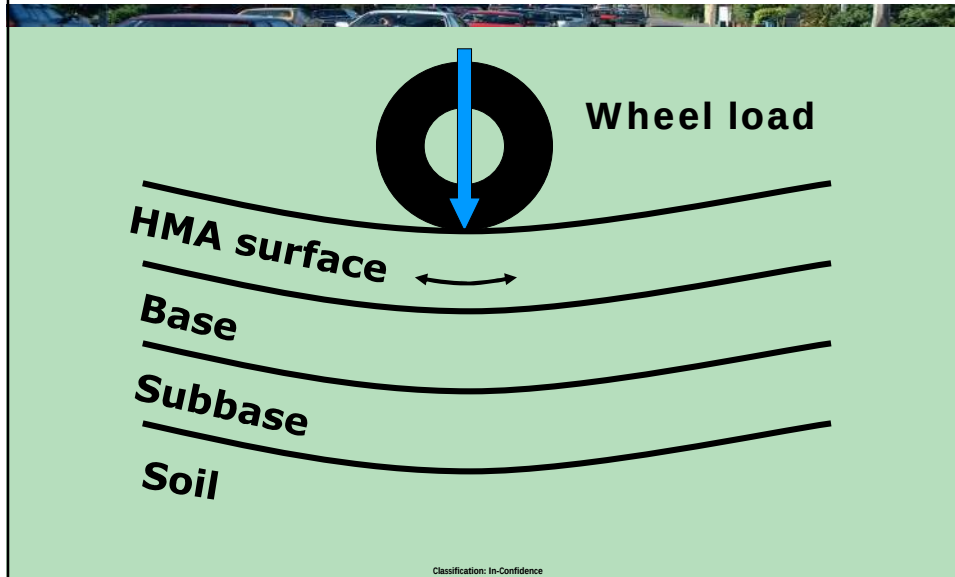


1. Load-related
2. Temperature-related
3. Moisture-related
4. Age-related

Classification: In-Confidence

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Load-Related: Fatigue



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Propagation of Fatigue Cracking



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Early Stage of Fatigue Cracking



Classification: In-Confidence

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Intermediate Stage of Fatigue Cracking



Classification: In-Confidence

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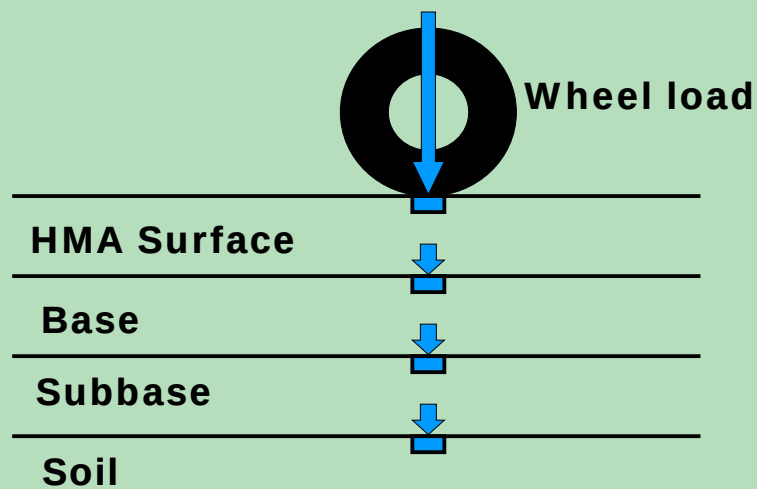
Advanced Stage of Fatigue Cracking



Classification: In-Confidence

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Load-Related: Permanent Deformation



Classification: In-Confidence

40

Minor Rutting



Classification: In-Confidence

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Severe Rutting



Classification: In-Confidence

42

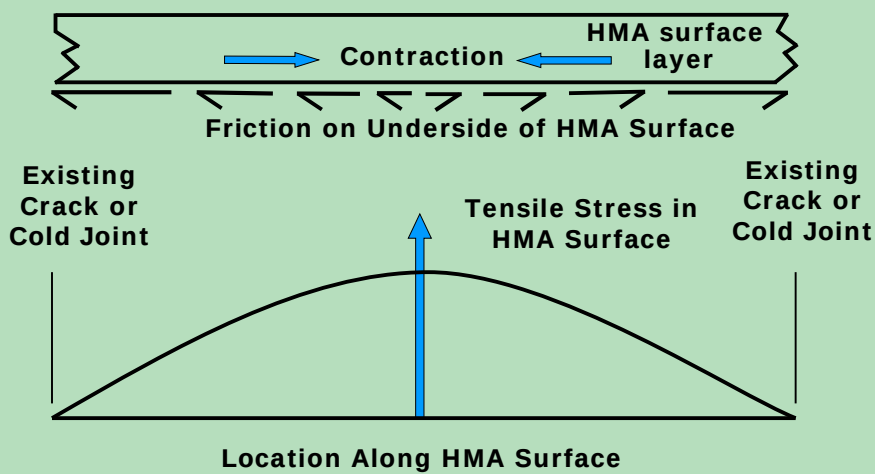
Rutting Confined to HMA Layer



Classification: In-Confidence

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Temperature-Related: Thermal Cracking



Classification: In-Confidence

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Thermal Cracking



Classification: In-Confidence

45

Thermal Cracks



Classification: In-Confidence

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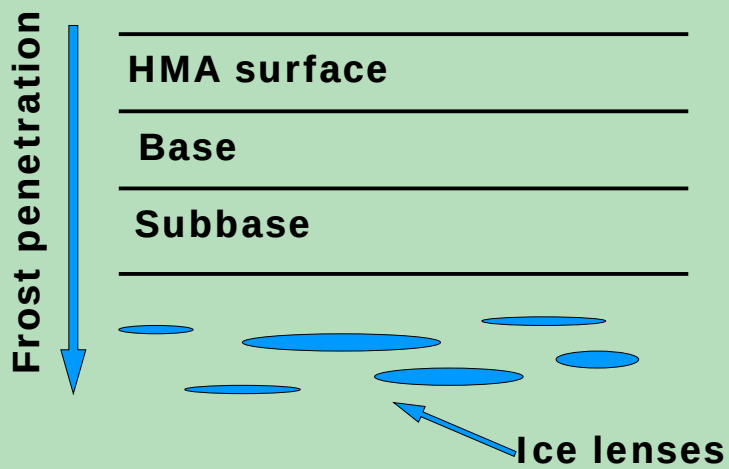
Wide Thermal Crack



Classification: In-Confidence

47

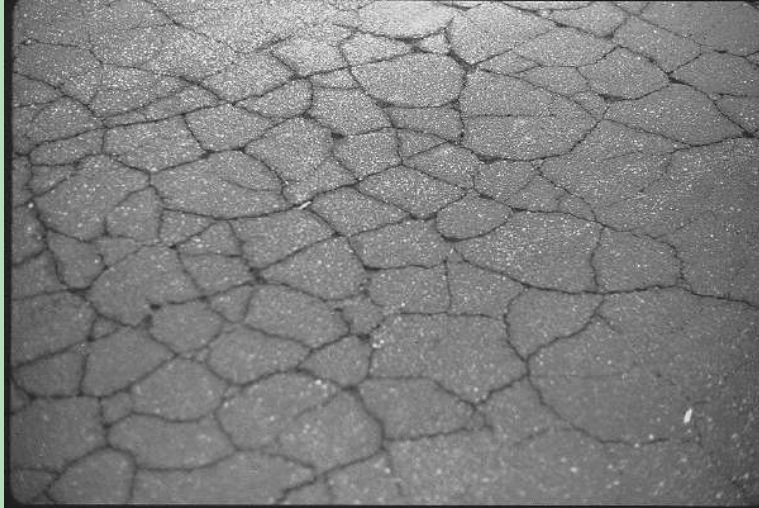
Temperature-Related: Frost Heave



Classification: In-Confidence

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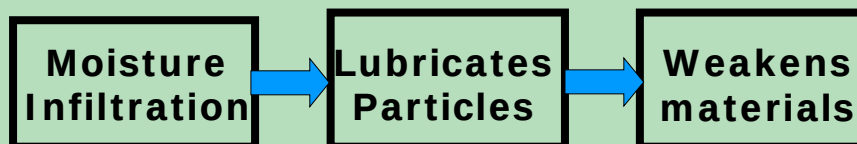
Thermal Cracks



Classification: In-Confidence

49

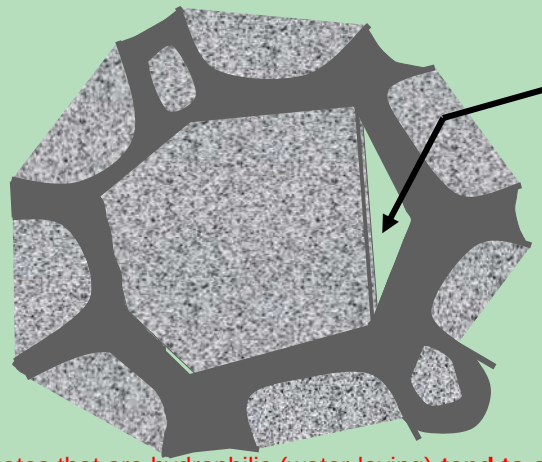
Moisture-Related: Strength Loss



Classification: In-Confidence

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Moisture-Related: Stripping



Separation
of bitumen
from
aggregate

Aggregates that are hydrophilic (water-loving) tend to strip more readily since water more easily replaces the bitumen film over each particle..

Classification: In-Confidence

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Stripping



Classification: In-Confidence

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Christchurch is the Pothole Capital of NZ



<https://www.stuff.co.nz/the-press/news/100847641/Christchurch-the-pothole-capital-of-new-zealand>

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Potholes



Classification: In-Confidence

54

Stripping Below Thermal Crack



Classification: In-Confidence

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Ageing Related Distress: Oxidized HMA Surface Layer



Classification: In-Confidence

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Rigid Pavements

§ Components

- Surface - PCC
- Base / Subbase
- Reinforcement
- Joints (configuration)
- Load transfer devices

Classification: In-Confidence

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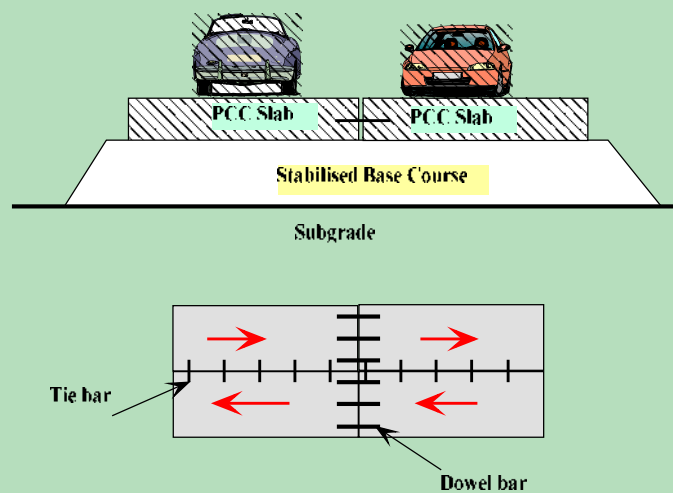
58



Classification: In-Confidence

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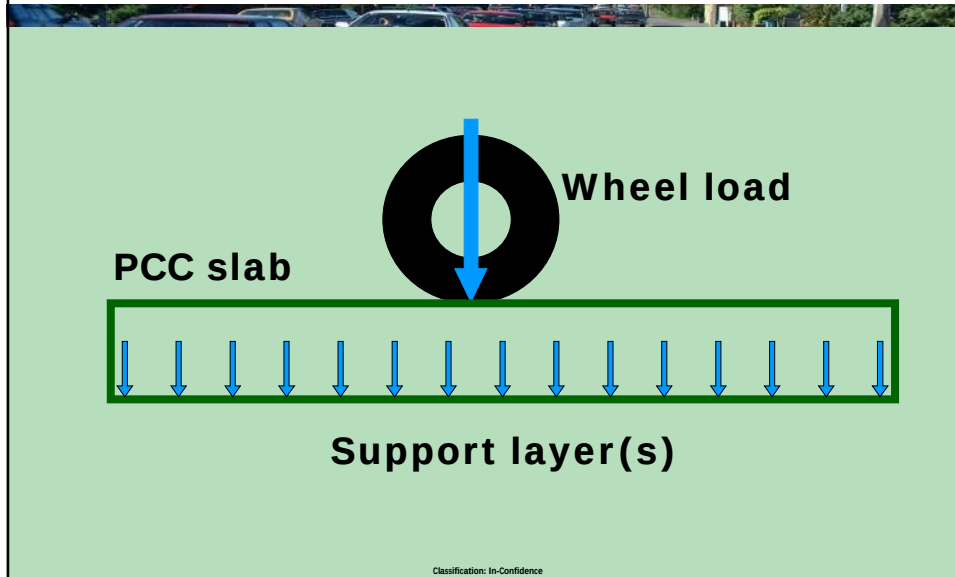
Rigid Pavements



Classification: In-Confidence

60

Distribution of Wheel Load



61

The Importance of the Base in The Rigid Pavements

- § Minimize or eliminate pumping.
- § Facilitate drainage of the surface water.
- § Provide control of frost action.
- § Expedition of construction.
- § Control of shrinkage and swell.

Classification: In-Confidence

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Basic Distress Mechanisms

§ Load-related

- Fatigue
- Faulting

§ Temperature-related

- Low-temp. mid-slab cracking
- High-temp. joint / crack distress

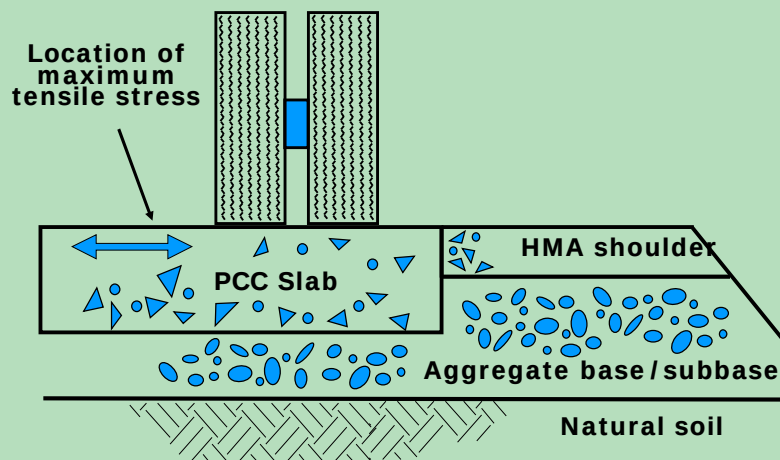
§ Moisture-related

- Pumping
- D - Cracking

Classification: In-Confidence

63

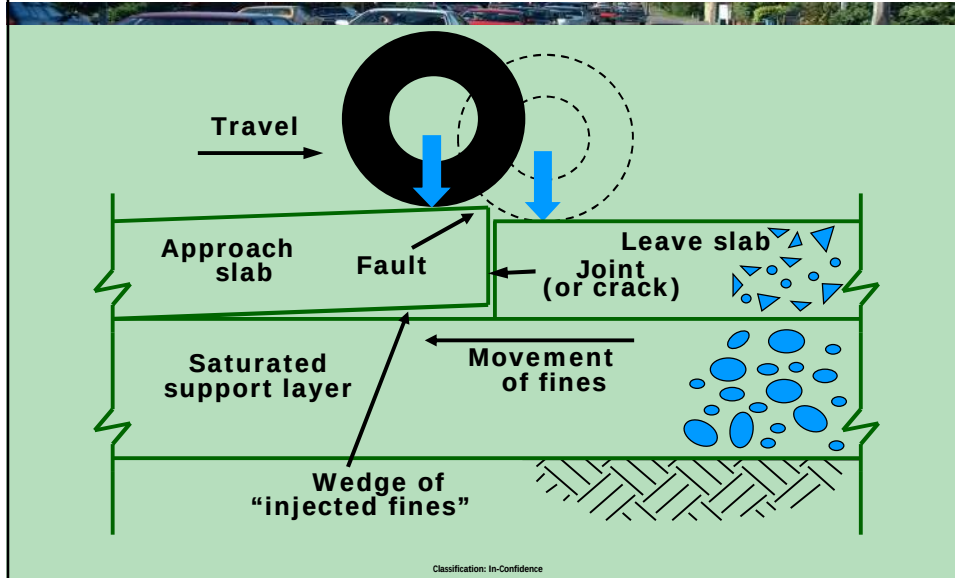
Load-Related: Fatigue



Classification: In-Confidence

64

Load-Related: Faulting



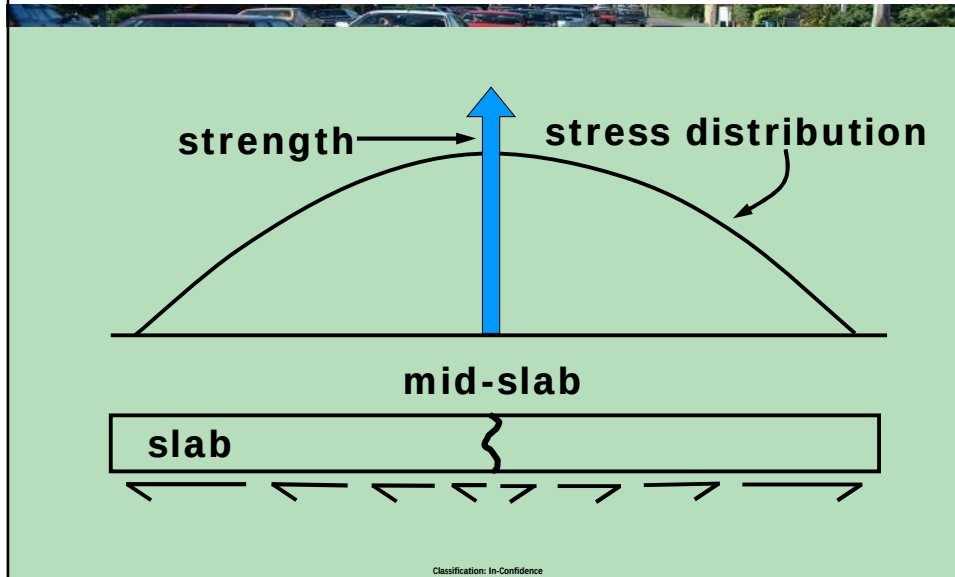
65

Load-Related: Faulting



66

Temperature-Related: Mid-Slab Cracking



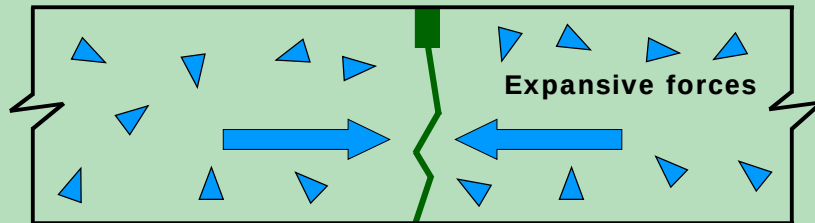
67

Temperature-Related: Mid-Slab Cracking



68

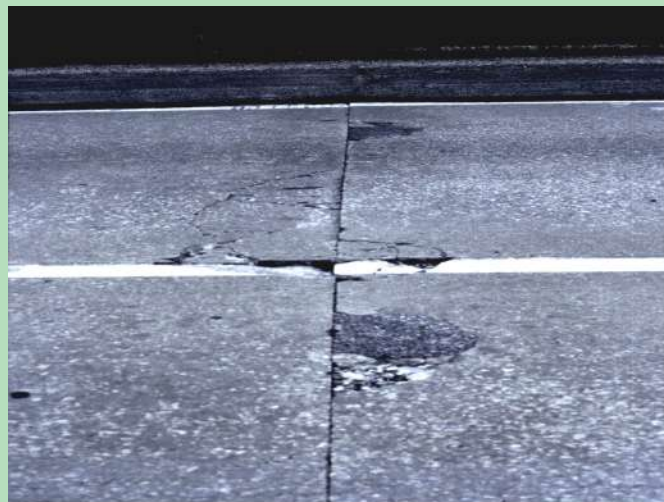
Temperature-Related: High-Temperature Joint / Crack Distress



Classification: In-Confidence

69

Temperature-Related: Spalling



Classification: In-Confidence

70

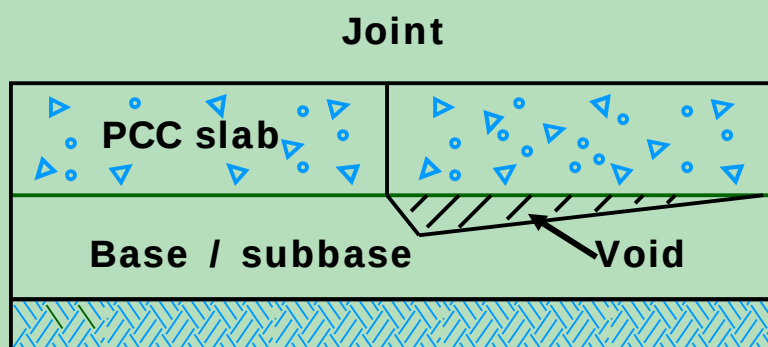
Temperature-Related: Blowup



Classification: In-Confidence

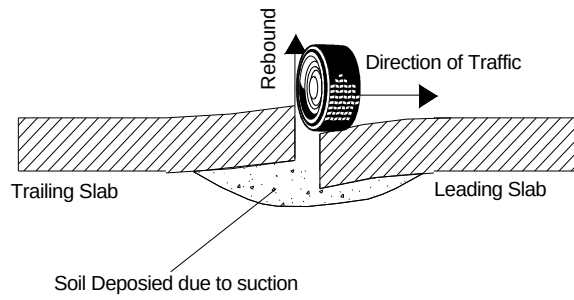
71

Moisture-Related: Pumping



Classification: In-Confidence

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Pumping of Rigid Pavements

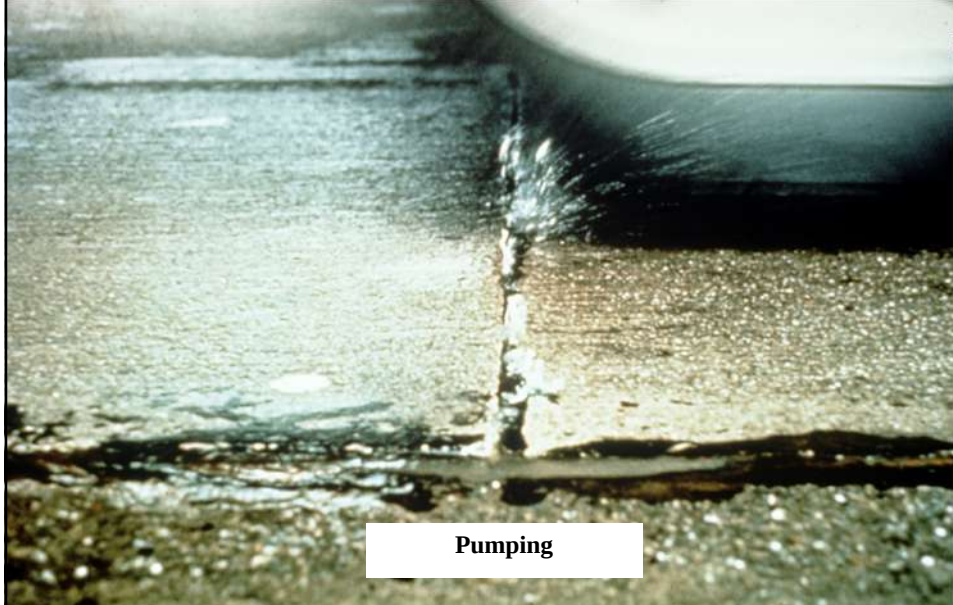
Classification: In-Confidence

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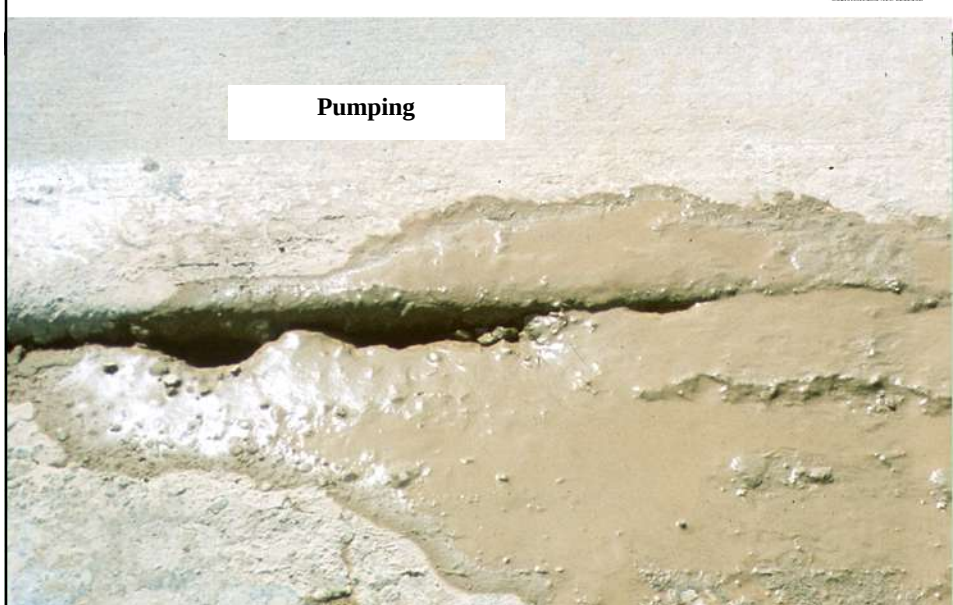
74

Pumping



Pumping

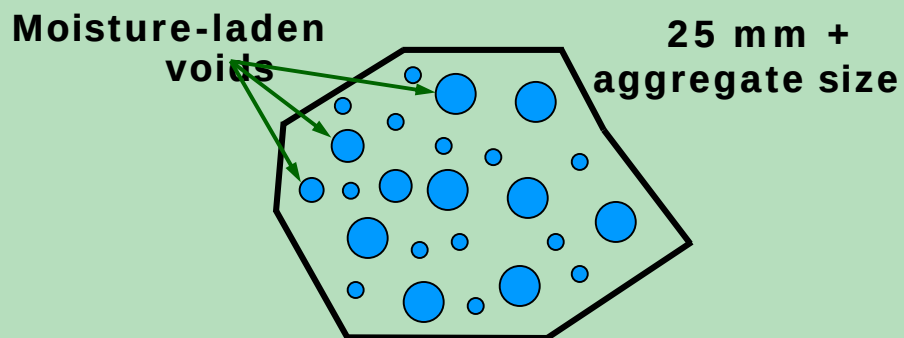
75



Pumping

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Moisture-Related: D-Cracking



Classification: In-Confidence

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Moisture-Related: D-Cracking



Classification: In-Confidence

78

Alkali-Silica Reactivity (ASR) Damage



Classification: In-Confidence

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Types of Concrete Pavements

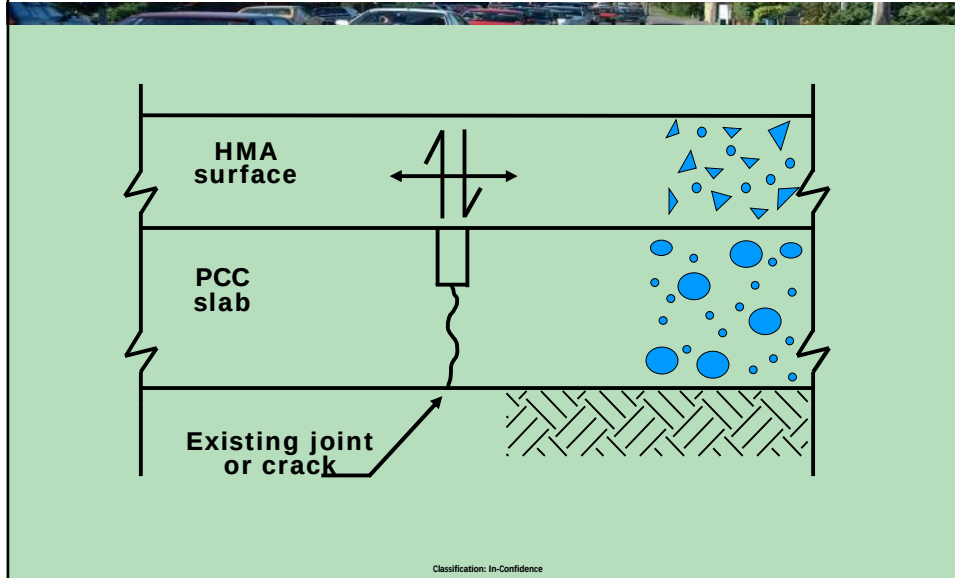


- § Jointed Plain Concrete Pavements (JPCP) (4.0-8.0 m).
- § Jointed Reinforced Concrete Pavements (JRCP) (8.0 -30 m).
- § Continuous Reinforced Concrete Pavements (CRCP).
- § Prestressed Concrete Pavements (PCP) (90-200 m).

Classification: In-Confidence

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Asphalt / Concrete Composite Pavements



81

Asphalt / Concrete Composite



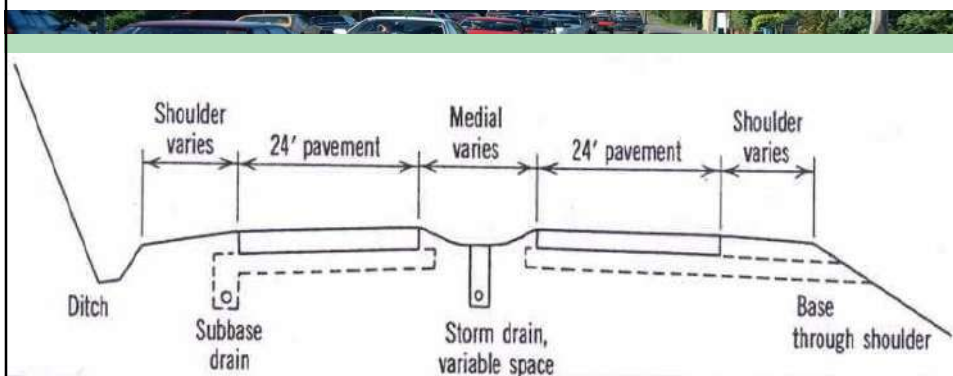
82

Reflection Cracks



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Highway vs. Airport Pavements

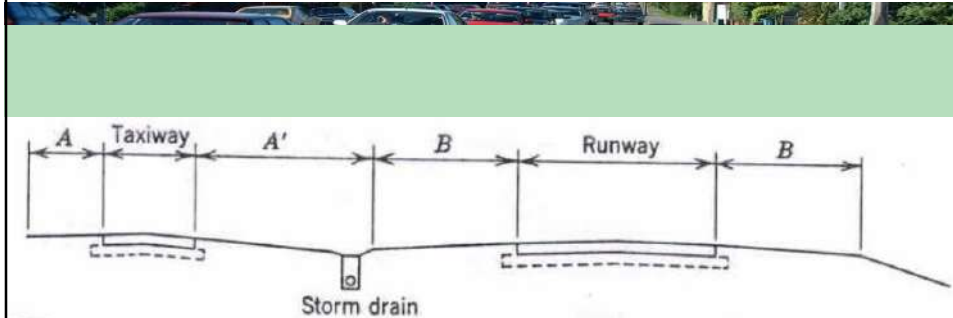


Typical Cross Section of a Highway

Classification: In-Confidence

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Highway vs. Airport Pavements



Typical Cross Section of Runway & Taxiway

Classification: In-Confidence

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Highway vs. Airport Pavements

1. Load magnitude
 - Hwys: up to 100 kN/axle
 - Arpts: up to 400 kN/wheel
2. Load repetition
 - Hwys: 1,000-2,000 trucks/day
 - Arpts : 20,000-40,000 coverages/life

Classification: In-Confidence

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Highway vs. Airport Pavements



3. Tyre Pressure

- Hwys: up to 830 kPa
- Arpts: up to 3 MPa

4. Load Distribution

- Hwys: Edge loading
- Arpts: Centre loading

Classification: In-Confidence

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Highway vs. Airport Pavements

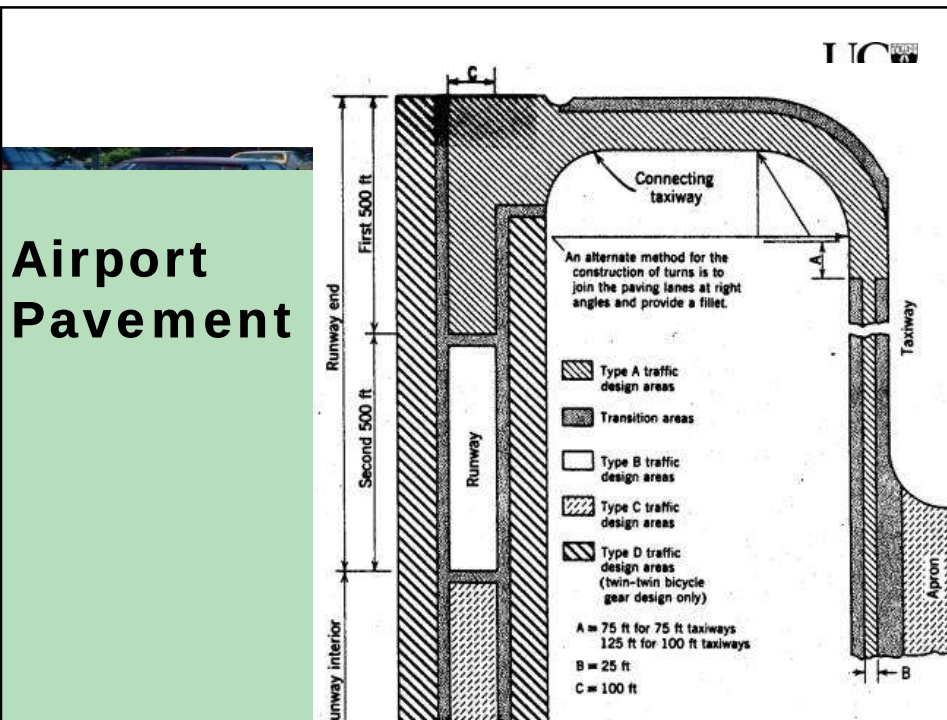


5. Load Type

- Hwys: Axle loads (single, tandem, tridem (Triaxles) or quad axles)
- Arpts: Gear loads (main and nose gears)

Classification: In-Confidence

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Airport Pavement

§ Runways, taxiways, and apron

§ Traffic areas A, B, and C

§ Traffic area A: Most critical
B: Less critical
C: Least critical

Classification: In-Confidence

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Airport Pavement



§ Takeoff is more critical than landing

§ At taking off

- Plane full of fuel
- Vibration
- Plane is not air lifted

Classification: In-Confidence

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Effect of Aircraft Wander

§ During landing and taking off on runway

- Large speed and large wander
- More pavement damage at centre

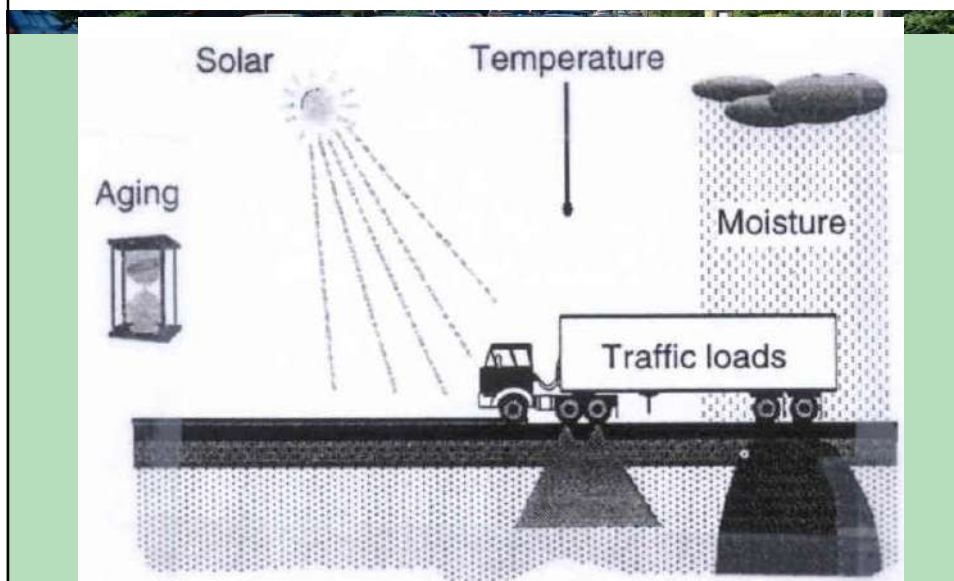
§ During taxiing

- Less speed and less wander
- More damage under main gears

Classification: In-Confidence

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Factors Affecting Pavement Performance



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Factors Affecting Pavement Design and Performance



- v Traffic Volume and Loading
- v Environmental Conditions
- v Material Properties
- v Construction and Maintenance
- v Failure Criteria

Classification: In-Confidence

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Traffic



Traffic has a major effect on pavement performance

- Ø Traffic volume
- Ø Traffic load
- Ø Tyre pressure
- Ø Rate of applying load

Classification: In-Confidence

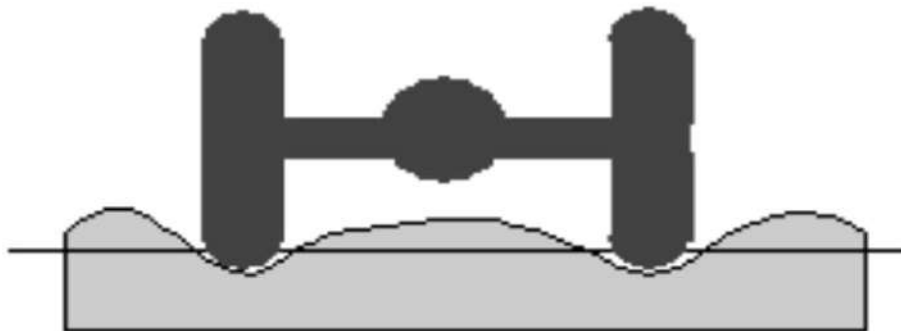
98



Heavier and slow speed trucks have worse effect

Classification: In-Confidence

99



**Larger tyre pressure is
more harmful to pavement**

Classification: In-Confidence

100

Super Single Tyres



Classification: In-Confidence

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Australian for “truck”



“Road Train”



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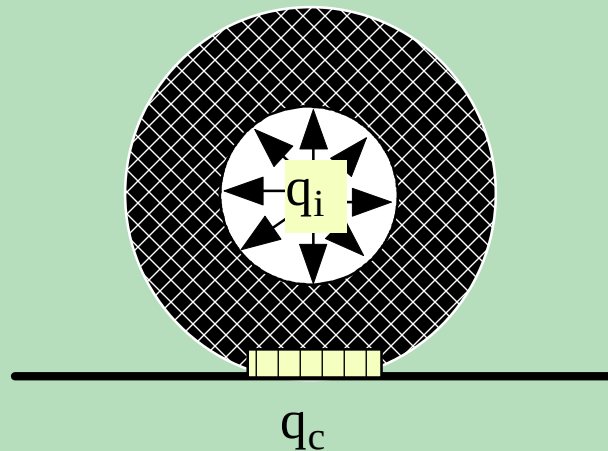
Traffic Volume and Loading

- v Axle Load and Axle Configuration
- v Standard Axle Load (80 kN), (18000 lb), (8.2 ton)
- v Number of Load Repetitions
- v Contact Area

Classification: In-Confidence

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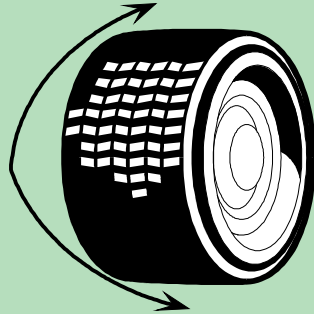
Contact Pressure and Tyre Inflation Pressure



Classification: In-Confidence

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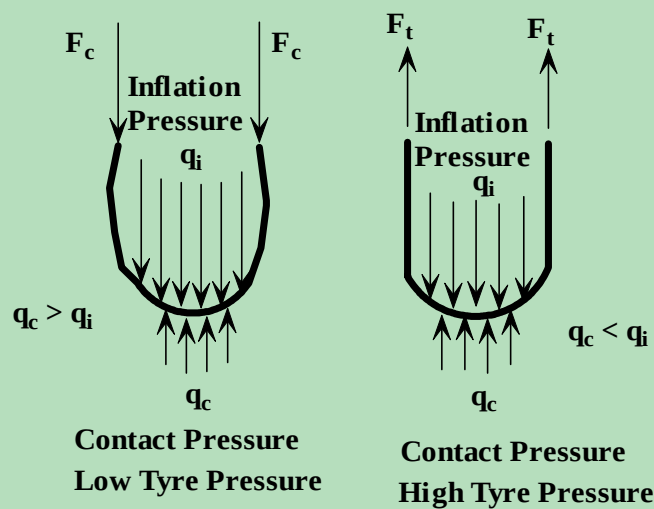
High Tyre Pressure Causes Tension in the Tyre Walls



Classification: In-Confidence

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Tyre Pressure



Classification: In-Confidence

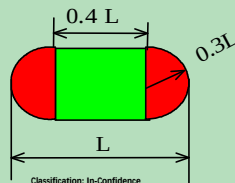
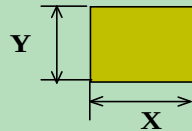
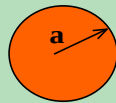
106

Contact Area

$$\text{Contact Area} = \frac{\text{Tyre Load}}{\text{Tyre Contact Pressure}} = \frac{P}{q_c}$$

$$\text{Contact area} = \pi * a^2 = \frac{P}{q_c}$$

$$a = \sqrt{\frac{P}{\pi * q_c}}$$



Classification: In-Confidence

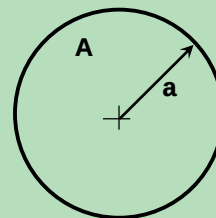
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Contact Area

- Assuming circular contact area (A) with a radius

$$A = \frac{\text{Load (P)}}{\text{Pressure (q)}}$$

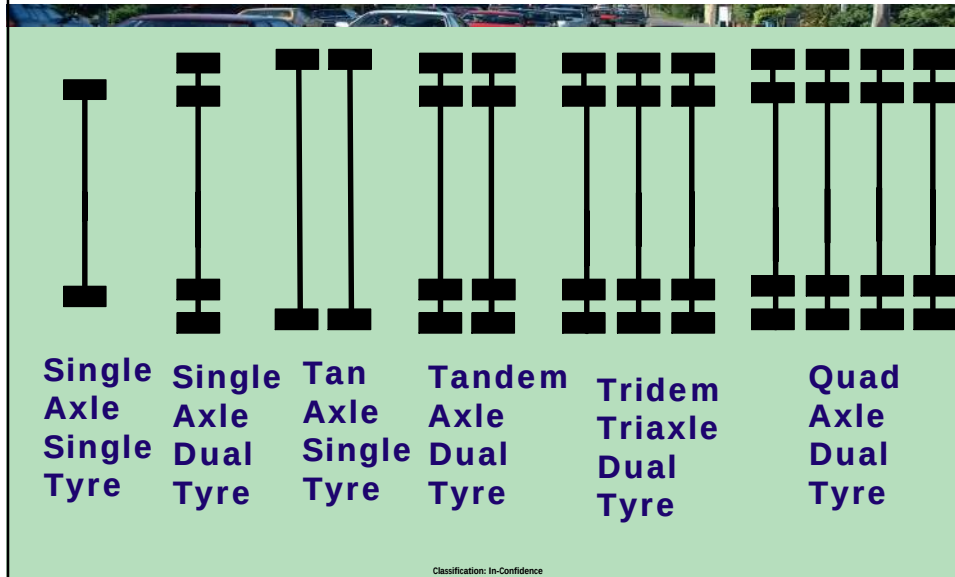
$$a = \sqrt{\frac{A}{\pi}} = \sqrt{\frac{P}{\pi * q}}$$



Classification: In-Confidence

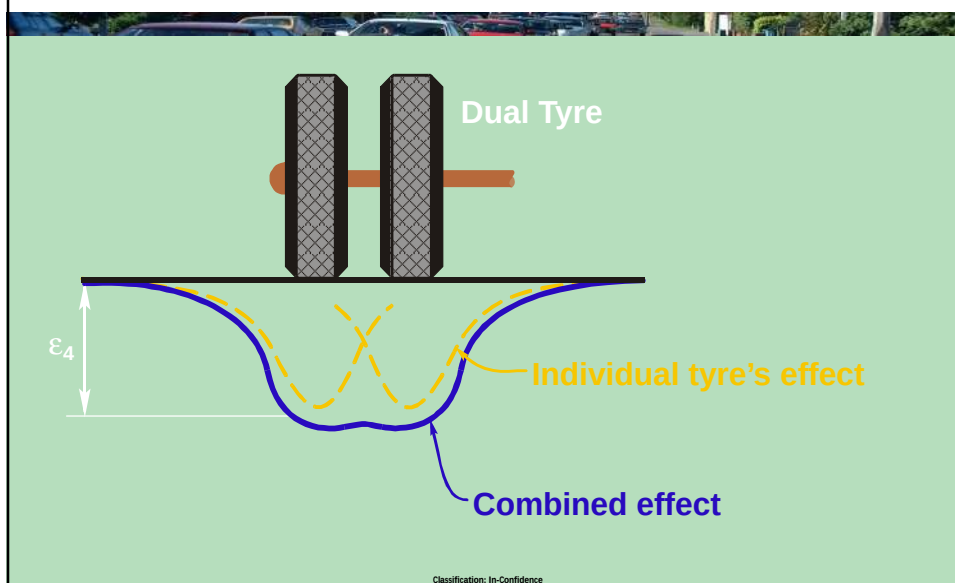
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Axle Configuration



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Dual tyre – Stresses Overlap



110

Tyre Pressure & Contact Pressure

§ Contact pressure is not constant throughout the contact area

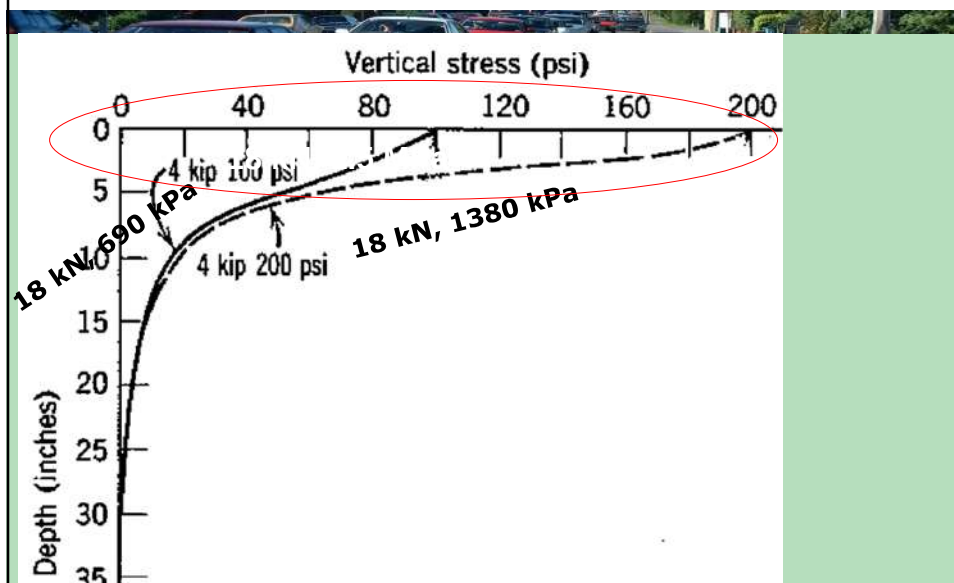
§ For simplicity we usually assume:

- Constant contact pressure
- Contact pressure (p_c) = Tyre pressure (p_t) = p

Classification: In-Confidence

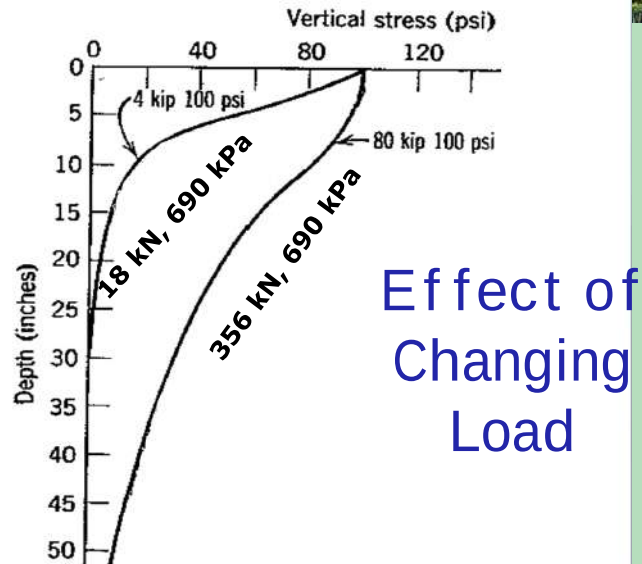
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Effect of Changing Tyre Pressure



112

Effect of Axle Load



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Tyre Pressure versus Load

- § Changing tyre pressure affects upper layers
- § Changing axle load affects deeper layers
- § Required quality of surface material is mostly determined by tyre pressure
- § Required pavement thickness is mostly determined by axle load magnitude

Classification: In-Confidence

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AXLE LOAD & REPETITIONS



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Vehicle Speed

§ Due to viscoelasticity, Asphalt materials properties depends on the rate of loading and temperature. The higher the speed, the faster the rate of loading and the higher is the stiffness (Dynamic modulus or resilient modulus) and vice versa. At very slow speed, the asphalt stiffness is low and this can explain the rutting and shoving developed in the pavement at parking lots and streets intersection. So the design speed should properly selected in the pavement design.

Classification: In-Confidence

116

Environmental Factors

§ Moisture

- Affect resilient modulus of base and subgrade
- Affect warping of concrete pavements

§ Temperature

- Affect Resilient Modulus for flexible pavement.
- Affect curling stress of concrete pavements.

§ Ageing

Classification: In-Confidence

117

Construction and maintenance

- § Construction and maintenance practice largely affects performance

Classification: In-Confidence

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Properties Related to Flexible Pavement Only



- § If the viscoelasticity of the material is to be considered, the creep compliance which is the reciprocal of the elastic moduli at various loading time should be specified.
- § The fatigue, rutting, and thermal cracking parameters are very important parameters in the design of flexible pavements.

Classification: In-Confidence

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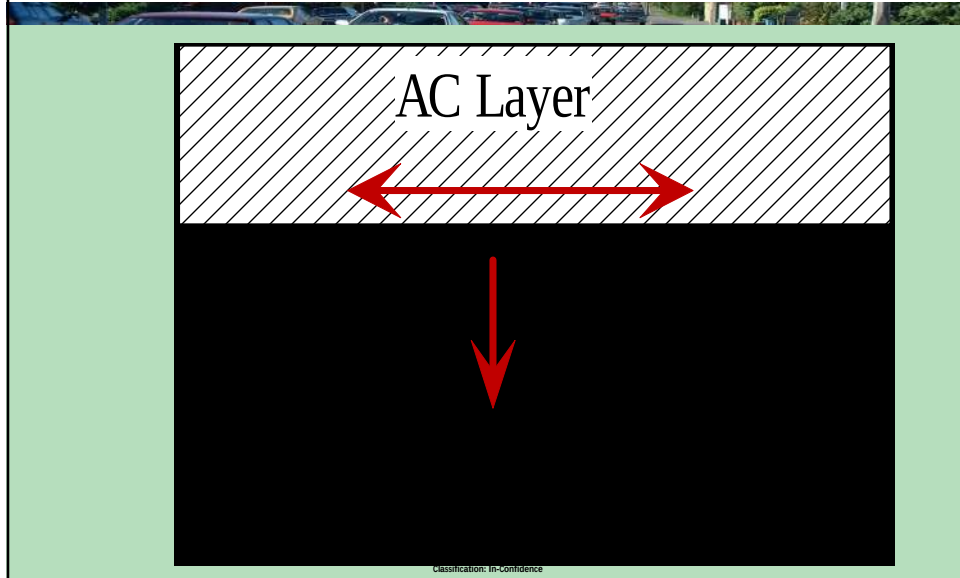
Failure Criteria



- § The failure criteria defines the conditions at which the pavement is considered at failure. It usually relates certain mode of failure (fatigue cracking, rutting, thermal cracking), pavement response (compressive strain, horizontal tensile strain...etc.) and the allowable number of load repetitions
- § **Fatigue:** The maximum horizontal tensile strain developed at the bottom of the asphalt layer is considered the main critical response for fatigue cracking.

Classification: In-Confidence

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Fatigue

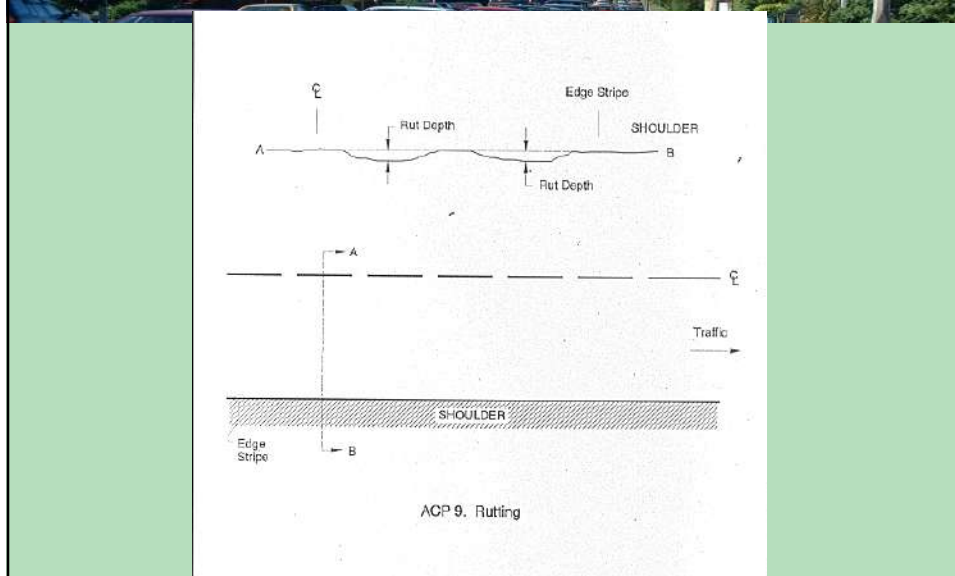


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§ **Rutting:** In several mechanistic design methods, rutting was attributed to the vertical compressive strain at the top of the subgrade. Models were developed to limit the vertical compressive strain at the top of the subgrade.

Classification: In-Confidence

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Rutting



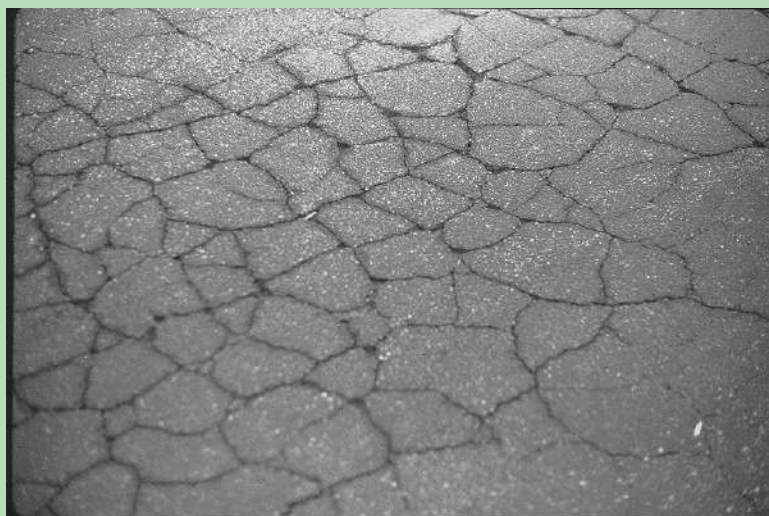
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§ **Thermal Cracking:** This type of distress is related to the change of HMA modulus due to the change of temperature. When the temperature drops down the modulus of asphalt layer increases and the material becomes more brittle and easy to crack.

Classification: In-Confidence

127

Thermal Cracks



Classification: In-Confidence

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Rigid Pavements

§ The most common failure criteria used in rigid pavement design is the fatigue failure criterion. Recently, pumping and erosion were considered as a failure criterion.

§ **Fatigue Cracking in Rigid Pavements**

§ The allowable number of load repetitions depends on the stress ratio. (If the stress ratio is less than 0.45 the slab can sustain infinite number of load repetitions without any fatigue failure.

Classification: In-Confidence

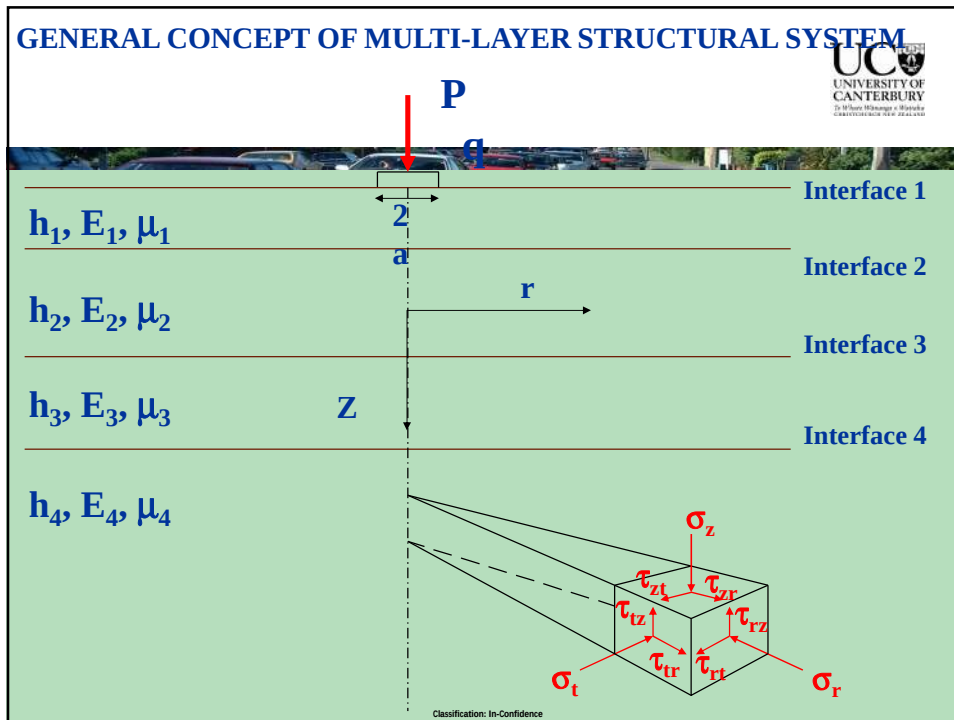
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SIMPLE EMPIRICAL MODELS

**SOPHISTICATED MECHANISTIC MODELS
BASED ON MATERIAL STRUCTURAL
BEHAVIOR**



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STRESSES ON EACH LAYER

UC
UNIVERSITY OF
CANTERBURY
To Whom It May Concern

3 NORMAL STRESSES : σ_z σ_r σ_t

6 SHEAR STRESSES : τ_{rt} τ_{tr} τ_{rz} τ_{zr} τ_{tz} τ_{zt}

DEFORMATION EQUATIONS:

$$\epsilon_z = \frac{1}{E} [\sigma_z - \mu(\sigma_r + \sigma_t)]$$

$$\epsilon_r = \frac{1}{E} [\sigma_r - \mu(\sigma_t + \sigma_z)]$$

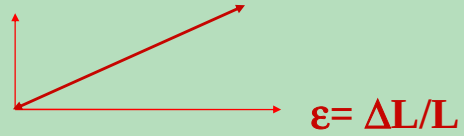
$$\epsilon_t = \frac{1}{E} [\sigma_t - \mu(\sigma_r + \sigma_z)]$$

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ELASTIC BEHAVIOUR

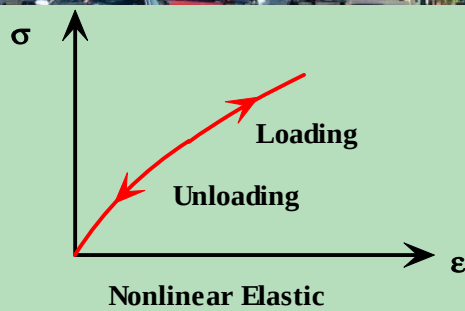


$$\sigma = P/A$$



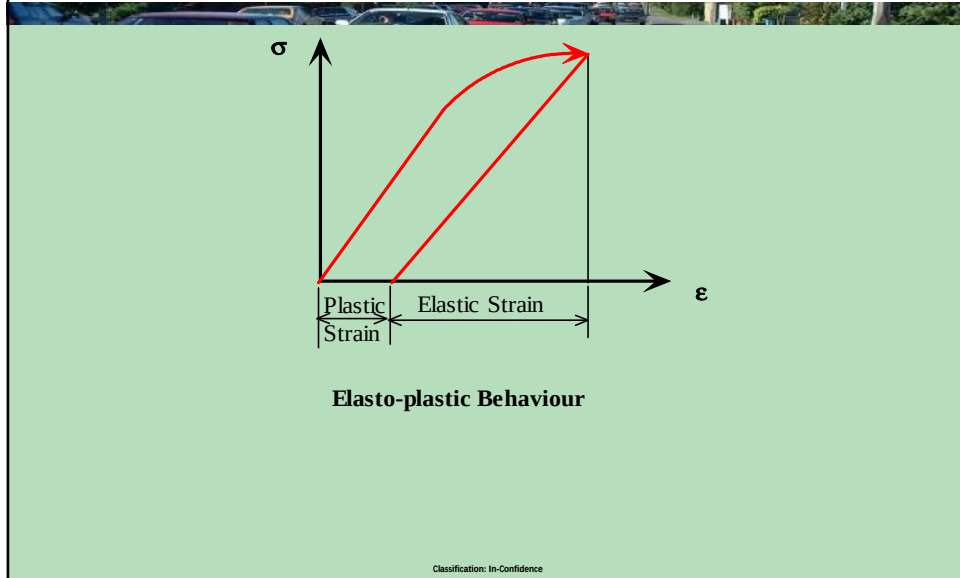
Classification: In-Confidence

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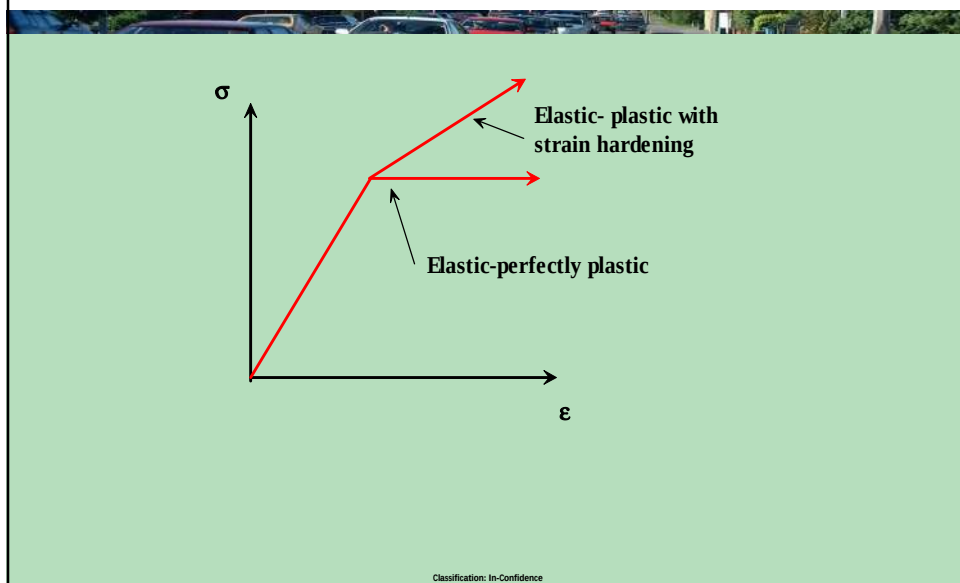


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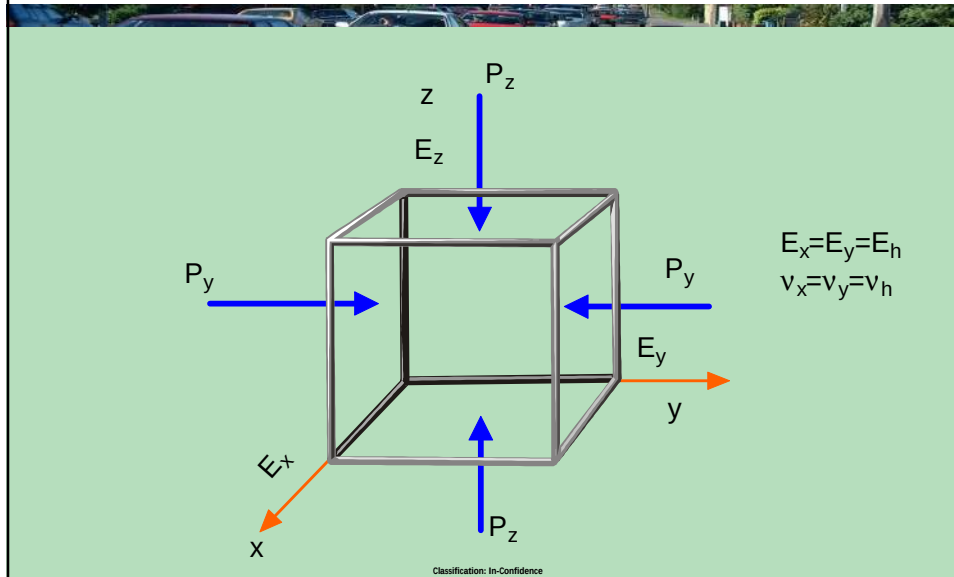
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Software Based on Elastic Layer Theory

CHEV5L or LAYER (CHEVRON)

BISAR (SHELL)

ELSYM5 (FHWA)

CIRCLY7

EVERSTRESS

WinJulea

Kenlayer

Classification: In-Confidence

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